MIL-B-43049C(IH)
30 September 1975
SUPERSEDING
MIL-B-0043049B(IH)
28 August 1970

MILITARY SPECIFICATION

BACKING PLATES; INSIGNIA, BRANCH OF SERVICE AND CAP, INFANTRY, ARMY

1. SCOPE

1.1 Scope.- This specification covers the plastic backing plates used by Army Infantry enlisted personnel.

1.2 Classification.- Backing plates shall be of the following types (see 6.2).

Type I - Backing Plate; Insignia, Service Cap, Infantry Blue
Type II - Backing Plate; Insignia, Branch of Service, Collar, Infantry Blue

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Fiberboard
PPP-B-676 - Boxes, Set-Up, Paperboard
STANDARDS
FEDERAL
FED-STD-406 - Plastics: Methods of Testing
MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
DRAWINGS
THE INSTITUTE OF HERALDRY
B-2-51 - Backing Plate for Cap & Collar US Army Infantry

(Copies of the specifications, standards and drawing required by the contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

(A miniature copy of drawing B-2-51, identified as figure 1 is attached for information purposes only.)

/FSC 8455/

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.
U.S. Army Color Card, The Official Colors for Branches, Agencies, Etc.

(Application for copies of the above color card should be addressed to The Color Association of the United States Inc., 200 Madison Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Standard sample.- Unless otherwise specified (see 6.2), the contracting officer shall furnish a sample of the backing plate for matching color and finish only.

3.2 First article sample.- When specified (see 6.2), before production is commenced, one sample of the finished backing plate from each cavity of the molds shall be furnished or made available to the contracting officer or his authorized representative for inspection and/or testing as specified in 4.2. Approval of the first article sample authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all provisions of the specification. The sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.3 Materials.- Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Plastic.- The backing plate shall be molded from virgin plastic material except the cellulose nitrate or cellulose acetate butyrate shall not be used.

3.4 Construction.- The backing plate, type I or type II shall be molded in accordance with the details and dimensions of drawing B-2-51. The backing plates shall meet the performance requirements specified herein.

3.4.1 Stability.- When tested as specified in 4.4.1, the backing plate shall not twist, chip, crack, break or blister and shall be free from warp. When the backing plate is placed on a surface plate with the beveled edge upward, at least 75 percent of the perimeter of the backing plate shall be in contact with the plate and there shall be no gap exceeding 1/32 inch.

3.4.2 Water absorption.- When tested as specified in 4.4.2, the water absorption shall not be greater than 2.5 percent.

3.4.3 Color.- The color shall match the color of the standard sample (see 3.1). When no standard sample is furnished, the color of the type I and type II backing plates shall be light blue approximating Cable No. 65014 (see 3.1). When tested as specified in 4.4.3, there shall be not noticeable change in color.

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3.4.4 Knockout pin mark.- Knockout pin marks and breakoff marks shall be trimmed flush within 1/64 inch of adjacent areas. There shall be no sharp edges. The marks shall not located in areas which are visable under actual use.

3.5 Finish.- The backing plates shall be smooth and the beveled section shall have a high uniform luster as provided by molding. A high luster is not required for all other areas.

3.6 Workmanship.- The backing plates shall be free of warp, twist, cracks, blisters, scratches, sink marks, flash, fins and roughness at breakoff point and shall meet the quality provisions as specified herein. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 First article sample inspection.- When first article samples are required, the completely finished item shall be inspected for all provisions of this specification applicable to end product examination and tests as required.

4.3 Inspection for acceptance.- Inspection for acceptance shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3.1 Component testing.-

4.3.1.1 Certification of complance.- The plastic material may be accepted on the contractor's certification of compliance that it meets the requirements of 3.3.1.

4.3.2 End item examination.- Defects found during the examination shall be classified in accordance with 4.3.2.1 and 4.3.2.2. The lots shall be examined in accordance with 4.3.2.3. The lot shall consist of all completely finished backing plates of one type offered for inspection at one time. The sample unit for these examinations shall be one completely finished backing plate.

4.3.2.1 Examination of the backing plate for visual defects.-

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Examination	Defect
Color and finish	Does not match standard sample or approximate Cable No. 65014.
	Not smooth.
	Not uniform shade, e.g., faded.
	Beveled section does not have a high uniform luster.
	Discoloration such as burn, spot, stain, streak or speck clearly noticeable.
	Any imbedded foreign matter.
	Any crack, break, blister, lump, pit, crater or other imperfection clearly noticeable.
	Any scratch on beveled surface.
Workmanship	Any chip, nick, dent, dig, gouge, sink mark, short, flash, fin, roughness or burr clearly noticeable.
	Any warp, twist, unevenness or distortion.
	Any hole not formed as indicated on drawing.
	Knockout pin or breakoff mark located in areas visible when in actual use.
	Not trimmed within 1/64 inch of adjacent areas.
	Edge of hole not clean or not smooth, e.g., any burr or flash fin.
	Breakoff mark not neatly accomplished, i.e., sharp or rough projection which is clearly noticeable or could possibly abrade cloth material on which worn.

4.3.2.2 Examination of backing plates for defects in dimensions.- Any dimension that is not within the specified tolerances shall be classified as a defect. The sample shall be one completely finished backing plate from each mold cavity.

4.3.2.3 AQL's and inspection levels.- The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection level	AQL Total
For defects applicable to 4.3.2.1	II (one class)	4.0
For defects applicable to 4.3.2.2	S-1 (one class)	4.0

4.3.3 Examination for count of backing plates in intermediate containers.- Backing plates packaged for shipment shall be examined to determine compliance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than specified or marked quantity of backing plates shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units. The lot size shall be the number of intermediate containers.

4.3.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table I. The sample unit shall be one shipping container fully prepared

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for delivery with the exception that it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE I Classification of preparation for delivery defects.-

Examination	Defects
Markings (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Net weight exceeds requirements.

4.3.5 End item testing.- Testing of the completely fabricated backing plates shall be performed in accordance with table II for the characteristics shown therein. The sample unit shall be three completely fabricated backing plates of each type. One determination per sample unit shall be performed for each characteristic. The requirements are applicable to the individual unit. The inspection level shall be S-1 and the AQL shall be 4.0 defects per hundred units.

TABLE II End item testing.-

Characteristic	Rqm't para	Test method	Results reported as	
			Pass or fail	To nearest
Stability	3.4.1	4.4.1	x	
Water absorbtion	3.4.2	4.4.2		0.1%
Colorfastness to light	3.4.3	4.4.3	x	

4.4 Tests.-

4.4.1 Stability at high temperatures and humidity.- The backing plate shall be tested in accordance with method 6011, procedure III of FED-STD-406, and shall then be examined to determine compliance with 3.4.1.

4.4.2 Test for water absorption.- The backing plate shall be tested in accordance with method 7031, procedure A of FED-STD-406, and shall be examined to determine compliance with 3.4.2. Dimensional changes in the backing plate shall not be considered and reported.

4.4.3 Colorfastness to light.- Backing plate shall be tested in accordance with method 6031 of FED-STD-406, except the number of hours of exposure shall be 20 hours. After testing is complete the samples shall be examined to determine compliance with 3.4.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.- Four hundred (400) backing plates of one type only shall be arranged on edge in four rows and packaged in a snug-fitting set-up paperboard box conforming to type I, variety 1, class A, style 4 or type I, variety 1, class C, style 1 of PPP-B-676. Dividers made of the same material as the box shall be positioned vertically and horizontally between the rows. Box closure shall be in accordance with the appendix of the specification.

5.1.2 Level C.- Backing plates of one type only shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his own standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- Ten thousand (10,000) backing plates of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to V3s or V3c of PPP-B-636. Closure and reinforcement shall be in accordance with the appendix of the applicable specification.

5.2.2 Level B.- Ten thousand (10,000) backing plates of one type only shall be packed in a snug-fitting fiberboard shipping container conforming to class Domestic, F or SF, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.3 Level C.- Backing plates of one type only shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules and requlations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The type I and type II backing plates covered by this specification are intended for the use of enlisted Infantry personnel to mount their cap and branch of service insignia.

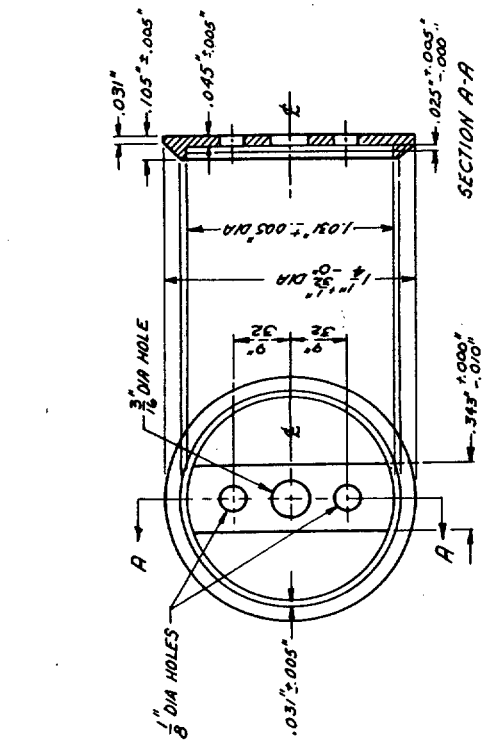
6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of backing plate required (see 1.2).
(c) When a standard sample will not be furnished (see 3.1).
(d) When first article is required (see 3.2).
(e) Selection of applicable levels of packaging and packing.

CUSTODIANS:	REVIEW ACTIVITIES:	USER ACTIVITIES:	PREPARING ACTIVITY:
Army	IH		The Institute of Heraldry,
Other		DP	United States Army

Project No. 8455-A382

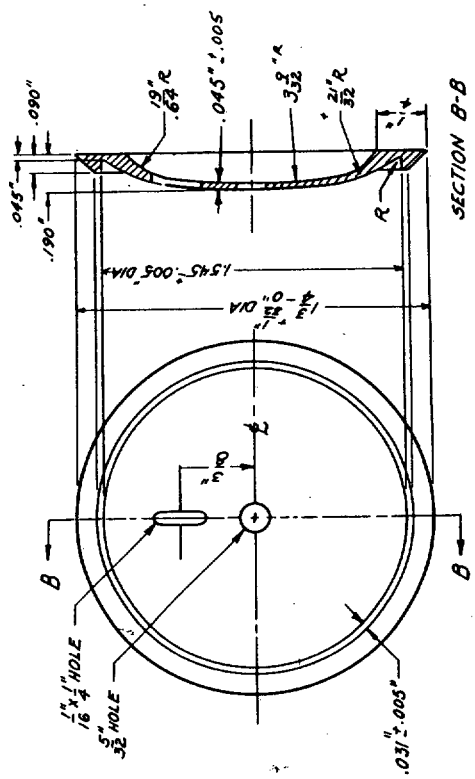
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TYPE II - BACKING PLATE, INSIGNIA, BRANCH OF SERVICE, COLLAR, INFANTRY BLUE

SUPERSEDES NATICK DWG Nº 9-1-157 DTD 28 DECEMBER 1961

FIGURE 1



TYPE I. BACKING PLATE, INSIGNIA, SERVICE CAP, INFANTRY BLUE

This is a miniature copy of THE INSTITUTE OF HERALDRY Drawing No. 8-2-5/, dated 26 May 1969 Revision dated 3 Sep 1975. This copy is for information only. Drawing referenced in para. 2.1 governs.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-B-43049C(IH) Backing Plates, Insg, BOS and Cap, Infantry, Army

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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